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MATERIALS SIMULATION PRACTICAL 1

Stationary Diffusion in Two Dimensions

Governing Equations and Finite-Difference Discretization

Report, Part I • Tasks 1 to 4

Author	Ali Ghelichkhani
E-mail	ali.ghelichkhani@fau.de
Supervisor	Xingchen Tan
Module	Foundations of Materials Simulation
Institute	Institute of Materials Simulation (WW8), FAU Erlangen-Nürnberg
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Abstract

We derive the partial differential equation that governs diffusion of a single species in a two-dimensional anisotropic medium and prepare it for numerical solution by finite differences. Combining Fick's first law with the continuity equation yields Fick's second law, a parabolic equation whose stationary limit is elliptic. In the principal axes of the diffusion tensor, the stationary equation reduces to $\partial^2_{xx}\rho + P\,\partial^2_{yy}\rho = 0$, where P is the ratio of the two principal diffusivities, and a rescaling of one coordinate transforms it into the Laplace equation. We discretize the equation with a second-order central-difference scheme on an equidistant mesh, introduce a row-wise node numbering, and incorporate Dirichlet conditions by elimination and Neumann conditions by ghost nodes that preserve second-order accuracy. The resulting linear system is sparse, pentadiagonal, block tridiagonal and irreducibly diagonally dominant; it is a nonsingular M-matrix whose only asymmetry stems from the Neumann rows and is removed by a diagonal scaling. These properties guarantee a unique discrete solution that is non-negative for physically admissible data and depends monotonically on them, and they prepare the implementation reported in Part II.

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1 Introduction

Diffusion is the net transport of matter that results from the random thermal motion of individual particles. Where the concentration of a species varies in space, more particles statistically jump from regions of high concentration into regions of low concentration than in the opposite direction, so that an initially inhomogeneous material homogenizes over time. Diffusion controls many processes in materials science, among them homogenization annealing, precipitation, oxidation and the motion of adsorbed atoms on surfaces. On the continuum scale it is described phenomenologically by Fick's two laws of diffusion [2, 3], which relate the diffusion flux to the concentration gradient and express the conservation of mass.

In this report we treat a two-dimensional domain on which a single species diffuses, for example adatoms on a crystal surface. We denote by $\rho(x, y)$ the concentration of the diffusing species, by $\boldsymbol{j}(x, y)$ the diffusion flux and by $\mathbf{D}$ the symmetric tensor of diffusion coefficients that connects the flux to the concentration gradient. Following the tutorial script [1], we derive and classify the governing equation (Section 2), discretize it by finite differences (Section 3), incorporate boundary conditions (Section 4) and assemble and analyse the resulting linear system (Section 5). Stationary heat conduction obeys an equation of identical structure, with temperature in place of concentration and the thermal conductivity tensor in place of $\mathbf{D}$, so all results apply to it as well. The numerical implementation and its verification (Task 5) form Part II and are provided as a Jupyter notebook. Each task of the script is restated, in paraphrased form, in a shaded box at the beginning of the section that answers it, and Table 1 gives an overview.

Table 1: Correspondence between the tasks of the tutorial script and this report.

Task	Content	Section
1.1	Fick's second law for anisotropic diffusion; classification	2.2, 2.3
1.2	Stationary diffusion in principal axes; meaning of P	2.4
1.3	Transformation to the Laplace equation	2.5
2.1	Second-order central-difference scheme	3.1
2.2	Node numbering and coefficient matrix	3.2, 3.3
3.1	Dirichlet boundary conditions	4.1
3.2	Neumann boundary conditions	4.2
4.1	Assembly of the linear system	5.1
4.2	Properties of the coefficient matrix	5.2

2 Governing equations

> **Task 1.1.** Using the continuity equation $\partial_t \rho + \nabla \cdot \boldsymbol{j} = 0$, derive Fick's second law in two dimensions for a material with anisotropic diffusion properties and without sources or sinks. Classify the resulting partial differential equation as elliptic, parabolic or hyperbolic.

2.1 Fick's first law for anisotropic media

Fick's first law states that the diffusion flux is proportional to the negative concentration gradient. In an isotropic medium the constant of proportionality is a scalar diffusion coefficient D, and the flux points exactly down the gradient, $\boldsymbol{j} = -D\nabla\rho$. In an anisotropic medium, such as a crystal surface with channels along which atoms jump more easily, the mobility depends on direction, and the scalar becomes a second-rank tensor:

$$\boldsymbol{j} = -\mathbf{D}\,\nabla\rho, \qquad \mathbf{D} = \begin{pmatrix} D_{xx} & D_{xy} \\ D_{xy} & D_{yy} \end{pmatrix}. \tag{2.1}$$

In components,

$$j_x = -\left(D_{xx}\,\partial_x\rho + D_{xy}\,\partial_y\rho\right), \qquad j_y = -\left(D_{xy}\,\partial_x\rho + D_{yy}\,\partial_y\rho\right). \tag{2.2}$$

The tensor is symmetric by Onsager's reciprocal relations [4, 5]. It is also positive definite, because the second law of thermodynamics requires non-negative entropy production, which here demands that the flux has a positive component down the gradient, $-\boldsymbol{j}\cdot\nabla\rho = \nabla\rho^{\mathsf{T}}\mathbf{D}\,\nabla\rho > 0$ for every $\nabla\rho \neq \mathbf{0}$. Unless $\nabla\rho$ happens to be an eigenvector of $\mathbf{D}$, the flux in (2.2) is therefore not antiparallel to the gradient: the off-diagonal coefficient D_{xy} couples a gradient in y to a flux in x and vice versa.

2.2 Continuity equation and Fick's second law (Task 1.1)

Consider an arbitrary fixed control area V with boundary ∂V and outward unit normal $\boldsymbol{n}$. In the absence of sources and sinks, the amount of the species in V changes only through the flux across ∂V:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho\,\mathrm{d}A = -\oint_{\partial V} \boldsymbol{j}\cdot\boldsymbol{n}\,\mathrm{d}s = -\int_V \nabla\cdot\boldsymbol{j}\,\mathrm{d}A, \tag{2.3}$$

where the second equality is the divergence theorem. Because V is fixed, the time derivative can be taken inside the integral, which gives $\int_V (\partial_t\rho + \nabla\cdot\boldsymbol{j})\,\mathrm{d}A = 0$. Since V is arbitrary and the integrand is assumed continuous, the integrand vanishes pointwise. This yields the continuity equation

$$\partial_t\rho + \nabla\cdot\boldsymbol{j} = 0. \tag{2.4}$$

Inserting Fick's first law (2.1) into (2.4) gives Fick's second law for an anisotropic medium,

$$\partial_t\rho = \nabla\cdot(\mathbf{D}\,\nabla\rho). \tag{2.5}$$

For a homogeneous medium, $\mathbf{D}$ does not depend on position. Expanding the divergence and using $\partial_x\partial_y\rho = \partial_y\partial_x\rho$ then yields

$$\boxed{\partial_t\rho = D_{xx}\,\partial^2_{xx}\rho + 2D_{xy}\,\partial^2_{xy}\rho + D_{yy}\,\partial^2_{yy}\rho.} \tag{2.6}$$

The mixed derivative is the signature of anisotropy in a coordinate system that is not aligned with the principal axes of $\mathbf{D}$. For a position-dependent tensor, additional first-order terms such as $(\partial_x D_{xx})\,\partial_x\rho$ appear, while the second-order part of the equation remains unchanged.

2.3 Classification (Task 1.1)

A linear second-order equation in the variables $\boldsymbol{z} = (z_1, \ldots, z_n)$ has the form

$$\sum_{k,l} a_{kl}\,\partial_{z_k}\partial_{z_l} u + (\text{lower-order terms}) = 0$$

with a symmetric coefficient matrix $\mathbf{a} = (a_{kl})$. Its type is determined by the signs of the eigenvalues of $\mathbf{a}$ [6]: the equation is elliptic if all eigenvalues are non-zero and of one sign, hyperbolic if one eigenvalue differs in sign from the others, and parabolic if exactly one eigenvalue vanishes, the others are of one sign and the corresponding variable enters through a first-order derivative.

For (2.6) in the variables (t, x, y), the coefficient matrix is

$$\mathbf{a} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & D_{xx} & D_{xy} \\ 0 & D_{xy} & D_{yy} \end{pmatrix}, \tag{2.7}$$

with eigenvalues 0, D_1 and D_2, where $D_1, D_2 > 0$ are the eigenvalues of the positive definite tensor $\mathbf{D}$. The zero eigenvalue belongs to t, which enters through the first derivative $\partial_t \rho$. Fick's second law is therefore *parabolic*. Restricted to the two spatial variables, the discriminant of the principal part is

$$(2D_{xy})^2 - 4D_{xx}D_{yy} = -4 \det \mathbf{D} < 0, \tag{2.8}$$

so the spatial operator, and hence the stationary equation of Section 2.4, is *elliptic*. The classification has direct physical meaning. The parabolic equation describes an irreversible, smoothing evolution in which a local perturbation is felt instantaneously everywhere; it requires an initial condition and boundary conditions. The elliptic stationary equation describes an equilibrium in which the solution at every point depends on the boundary data along the entire boundary; it requires boundary conditions only.

2.4 Stationary diffusion in principal axes (Task 1.2)

> **Task 1.2.** Show that stationary diffusion is governed by $\partial^2_{xx}\rho + P\,\partial^2_{yy}\rho = 0$. Explain the meaning of x, y and P, and how the tensor of diffusion coefficients determines P.

In the stationary state $\partial_t \rho = 0$, and (2.5) reduces to $\nabla \cdot (\mathbf{D}\,\nabla\rho) = 0$. We assume a homogeneous medium.

Assumption 1. The diffusion tensor $\mathbf{D}$ is symmetric, positive definite and independent of position.

By the spectral theorem, $\mathbf{D}$ has an orthonormal eigenbasis, that is

$$\mathbf{D} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}}, \qquad \mathbf{Q} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \mathbf{\Lambda} = \operatorname{diag}(D_1, D_2), \tag{2.9}$$

with principal diffusivities and rotation angle

$$D_{1,2} = \frac{D_{xx} + D_{yy}}{2} \pm \sqrt{\left(\frac{D_{xx} - D_{yy}}{2}\right)^2 + D_{xy}^2}, \qquad \tan 2\theta = \frac{2D_{xy}}{D_{xx} - D_{yy}}. \tag{2.10}$$

We introduce coordinates along the principal axes by $(x', y')^{\mathsf{T}} = \mathbf{Q}^{\mathsf{T}}(x, y)^{\mathsf{T}}$. By the chain rule the gradient transforms as $\nabla = \mathbf{Q}\,\nabla'$, where ∇' denotes the gradient with respect to (x', y'). Hence

$$\nabla \cdot (\mathbf{D}\,\nabla\rho) = \nabla'^{\mathsf{T}}\mathbf{Q}^{\mathsf{T}}\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}}\mathbf{Q}\,\nabla'\rho = \nabla'^{\mathsf{T}}\mathbf{\Lambda}\,\nabla'\rho = D_1\,\partial^2_{x'x'}\rho + D_2\,\partial^2_{y'y'}\rho, \tag{2.11}$$

and the mixed derivative has disappeared. Dividing by $D_1 > 0$ and dropping the primes gives the stationary diffusion equation in the form of the script,

$$\boxed{\partial^2_{xx}\rho(x,y) + P\,\partial^2_{yy}\rho(x,y) = 0, \qquad P := \frac{D_2}{D_1} > 0.} \tag{2.12}$$

Meaning of x, y and P. In (2.12), x and y are Cartesian coordinates along the principal axes of $\mathbf{D}$, that is, along its eigenvectors. In this frame each flux component depends only on the gradient along the same axis, $j_x = -D_1\,\partial_x\rho$ and $j_y = -D_2\,\partial_y\rho$. The parameter P is the ratio of the principal diffusivities and measures the degree of anisotropy: $P = 1$ for an isotropic medium, $P > 1$ if diffusion along y is faster, and $P < 1$ if it is slower. Exchanging the roles of the axes replaces P by $1/P$.

Influence of D on P. The parameter P depends on $\mathbf{D}$ only through its eigenvalues, which are invariant under rotation. The off-diagonal coefficient D_{xy} therefore affects P only through (2.10), while it determines the orientation θ of the principal axes. If the laboratory axes already coincide with the principal axes ($D_{xy} = 0$), then $P = D_{yy}/D_{xx}$. Equivalently, P is fixed by the two invariants of $\mathbf{D}$ through $(1+P)^2/P = (\operatorname{tr}\mathbf{D})^2/\det\mathbf{D}$. Crystal symmetry restricts $\mathbf{D}$ [5]: in a bulk cubic crystal diffusion is isotropic ($P = 1$), whereas on low-symmetry surfaces, such as fcc(110) with its close-packed atomic rows, adatoms often migrate much faster along the rows than across them, which corresponds to $P \gg 1$ or $P \ll 1$.

2.5 Transformation to the Laplace equation (Task 1.3)

> **Task 1.3.** Find a transformation of variables that converts the stationary equation into the Laplace equation $\partial^2_{\tilde{x}\tilde{x}}\tilde{\rho} + \partial^2_{\tilde{y}\tilde{y}}\tilde{\rho} = 0$.

We rescale the y coordinate and leave x unchanged,

$$\tilde{x} = x, \qquad \tilde{y} = \frac{y}{\sqrt{P}}, \qquad \tilde{\rho}(\tilde{x}, \tilde{y}) := \rho\big(\tilde{x}, \sqrt{P}\,\tilde{y}\big). \tag{2.13}$$

By the chain rule, $\partial_{\tilde{y}}\tilde{\rho} = \sqrt{P}\,\partial_y\rho$ and $\partial^2_{\tilde{y}\tilde{y}}\tilde{\rho} = P\,\partial^2_{yy}\rho$, while $\partial^2_{\tilde{x}\tilde{x}}\tilde{\rho} = \partial^2_{xx}\rho$. Substitution into (2.12) yields the Laplace equation

$$\boxed{\partial^2_{\tilde{x}\tilde{x}}\tilde{\rho}(\tilde{x}, \tilde{y}) + \partial^2_{\tilde{y}\tilde{y}}\tilde{\rho}(\tilde{x}, \tilde{y}) = 0.} \tag{2.14}$$

The symmetric choice $\tilde{x} = x/\sqrt{D_1}$, $\tilde{y} = y/\sqrt{D_2}$ works as well; (2.13) has the advantage that $\tilde{x}$ and $\tilde{y}$ retain the dimension of length. The transformation has a simple interpretation: stationary anisotropic diffusion is equivalent to isotropic diffusion on a domain that is compressed by the factor $\sqrt{P}$ along the fast direction. A rectangle $[0, l_x] \times [0, l_y]$ aligned with the principal axes maps onto the rectangle $[0, l_x] \times [0, l_y/\sqrt{P}]$. Dirichlet data are carried over pointwise, whereas prescribed normal derivatives on the boundaries $y = \text{const}$ are multiplied by $\sqrt{P}$. Consequently, the classical properties of harmonic functions, such as the mean-value property, the maximum principle and uniqueness of the Dirichlet problem [6], also hold for the anisotropic problem.

From here on we follow the script and consider the isotropic case $D_{xx} = D_{yy} =: D$, for which $P = 1$. We nevertheless keep P in the discretization, since it costs nothing and covers anisotropic media aligned with the mesh.

3 Finite-difference discretization

3.1 Central-difference scheme (Task 2.1)

> **Task 2.1.** Discretize the stationary equation with a second-order accurate central-difference scheme on an equidistant mesh with node spacings h_x and h_y. Denote the values at neighbouring nodes by indices such as $\rho_{i+1,j}$ for the right neighbour of node (i, j).

We consider the rectangle $[0, l_x] \times [0, l_y]$ and cover it with an equidistant mesh of $N_x \times N_y$ nodes,

$$\begin{aligned} x_i &= (i-1)\,h_x, \quad i = 1, \dots, N_x, & h_x &= \frac{l_x}{N_x - 1}, \\ y_j &= (j-1)\,h_y, \quad j = 1, \dots, N_y, & h_y &= \frac{l_y}{N_y - 1}, \end{aligned} \tag{3.1}$$

and denote by $\rho_{i,j}$ the approximation of $\rho(x_i, y_j)$. The index i increases to the right and j increases upwards. Taylor expansion about (x_i, y_j) gives

$$\rho(x_i \pm h_x, y_j) = \rho \pm h_x\,\partial_x\rho + \frac{h_x^2}{2}\,\partial^2_{xx}\rho \pm \frac{h_x^3}{6}\,\partial^3_x\rho + \frac{h_x^4}{24}\,\partial^4_x\rho + \mathcal{O}\big(h_x^5\big), \tag{3.2}$$

with all derivatives evaluated at (x_i, y_j). Adding the two expansions cancels the odd terms, and we obtain the central second difference

$$\delta^2_x\rho_{i,j} := \frac{\rho_{i+1,j} - 2\rho_{i,j} + \rho_{i-1,j}}{h_x^2} = \partial^2_{xx}\rho(x_i, y_j) + \frac{h_x^2}{12}\,\partial^4_x\rho(x_i, y_j) + \mathcal{O}\big(h_x^4\big), \tag{3.3}$$

and analogously $\delta_y^2 \rho_{i,j}$ in the y direction. Replacing the derivatives in (2.12) by these differences yields the discrete equation at node (i, j),

$$\frac{\rho_{i+1,j} - 2\rho_{i,j} + \rho_{i-1,j}}{h_x^2} + P\,\frac{\rho_{i,j+1} - 2\rho_{i,j} + \rho_{i,j-1}}{h_y^2} = 0. \tag{3.4}$$

Its local truncation error, obtained by inserting the exact solution into (3.4), is

$$\tau_{i,j} = \frac{h_x^2}{12}\,\partial_x^4 \rho + P\,\frac{h_y^2}{12}\,\partial_y^4 \rho + \mathcal{O}\big(h_x^4 + h_y^4\big) = \mathcal{O}\big(h_x^2 + h_y^2\big)\,, \tag{3.5}$$

so the scheme is second-order accurate [7]. Multiplying (3.4) by $-h_x^2$ gives the form used in the remainder of this report,

$$\boxed{2(1+\lambda)\,\rho_{i,j} - \rho_{i-1,j} - \rho_{i+1,j} - \lambda\,\rho_{i,j-1} - \lambda\,\rho_{i,j+1} = 0, \qquad \lambda := P\,\frac{h_x^2}{h_y^2}.} \tag{3.6}$$

The dimensionless weight λ combines the anisotropy ratio with the aspect ratio of the mesh cells. The five-point stencil is shown in Figure 1.

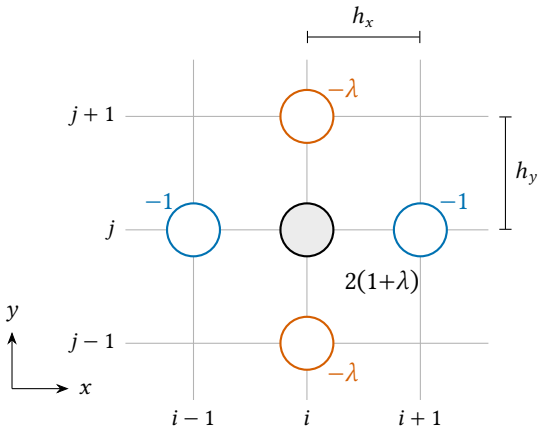


Figure 1: Five-point stencil of the central-difference scheme (3.6) at node (i, j) with the weights of the neighbouring nodes. For an isotropic medium on a square mesh, $\lambda = 1$.

Remark 3.1. For an isotropic medium on a square mesh ($P = 1$, $h_x = h_y$), we have $\lambda = 1$, and (3.6) states that $\rho_{i,j}$ equals the arithmetic mean of its four neighbours. This discrete mean-value property implies a discrete maximum principle: no interior node can exceed all of its neighbours, so extrema of the discrete solution lie on the boundary, exactly as for harmonic functions. Moreover, the truncation error (3.5) vanishes for polynomials of degree at most three in each variable, so that linear concentration profiles, as expected in the test problems of Part II, are reproduced exactly up to round-off.

3.2 Node numbering (Task 2.2a)

Task 2.2. A rectangular domain whose edges are aligned with the principal axes of the diffusion tensor is discretized by 5×5 nodes. Write the scheme of Task 2.1 as a linear system of equations.

(a) Number the nodes row-wise from the top left (node 1) to the bottom right (node 25), and sketch the mapping between the indices (i, j) and the node number for a general mesh of $N \times M$ nodes.

(b) Sketch the interior part of the coefficient matrix, whose entries follow from the factors of the nodal values in the scheme, and explain the difficulties that arise at the boundaries.

To write the discrete equations as a linear system, the two-dimensional node indices (i, j) must be mapped onto a single index k of a solution vector. Following the script, we number the nodes row-wise, starting at the top left node and ending at the bottom right node (Figure 2). For a mesh of $N \times M$ nodes, that is $N_x = N$ nodes per row and $N_y = M$ rows, the top row ($j = M$) receives the numbers 1 to N, the next row the numbers $N + 1$ to $2N$, and so forth. The mapping and its inverse are

$$k(i, j) = (M - j)\,N + i, \qquad i = 1 + \big((k-1) \bmod N\big), \qquad j = M - \left\lfloor \frac{k-1}{N} \right\rfloor, \tag{3.7}$$

with $k = 1, \dots, NM$. The four neighbours of node k follow directly from (3.7):

$$(i \pm 1, j) \;\mapsto\; k \pm 1, \qquad (i, j \pm 1) \;\mapsto\; k \mp N. \tag{3.8}$$

Horizontal neighbours are adjacent in the vector, and vertical neighbours are N positions apart. In a zero-based implementation with the row index $r = M - j$ counted from the top and $i' = i - 1$, the mapping reads $k' = rN + i'$. This is the row-major (C) ordering of NumPy, so that the solution vector can be reshaped directly into an $M \times N$ array whose first row is the top of the domain.

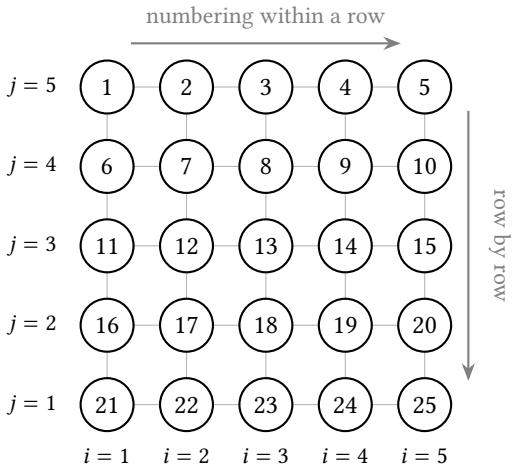


Figure 2: Row-wise numbering of a 5×5 mesh ($N = M = 5$) from the top left node ($k = 1$) to the bottom right node ($k = 25$). Node (i, j) receives the number $k = (M - j)N + i$; for example, the centre node $(3, 3)$ is node 13.

3.3 Coefficient matrix without boundary conditions (Task 2.2b)

Applying the stencil (3.6) to node k and using (3.8) gives the k-th row of the linear system,

$$-\lambda\,\rho_{k-N} - \rho_{k-1} + 2(1 + \lambda)\,\rho_k - \rho_{k+1} - \lambda\,\rho_{k+N} = 0. \tag{3.9}$$

Each row therefore contains at most five non-zero coefficients,

$$c_{k,k} = 2(1 + \lambda), \qquad c_{k,k\pm1} = -1, \qquad c_{k,k\pm N} = -\lambda, \tag{3.10}$$

located on the main diagonal, on the first sub- and superdiagonals and on the two diagonals at distance N. For the 5×5 mesh, the inner part of the matrix consists of the rows of the nine interior nodes $k \in \{7, 8, 9, 12, 13, 14, 17, 18, 19\}$; for instance, the row of the centre node reads

$$-\lambda\,\rho_8 - \rho_{12} + 2(1 + \lambda)\,\rho_{13} - \rho_{14} - \lambda\,\rho_{18} = 0. \tag{3.11}$$

Figure 3 shows the resulting sparsity pattern of the 25×25 coefficient matrix. The interior rows are complete, while the rows of the 16 boundary nodes cannot be formed from the stencil alone. Three difficulties arise in these rows.

1. *Missing neighbours.* At boundary nodes the stencil reaches nodes outside the mesh, with $i = 0$, $i = N + 1$, $j = 0$ or $j = M + 1$, which have no index. For the top and bottom rows the indices $k \mp N$ fall outside $\{1, \dots, NM\}$.
2. *Wrap-around of the numbering.* For a node on the right edge, $k + 1$ is the first node of the next row on the left edge (for example $5 \to 6$), and for a node on the left edge, $k - 1$ is the last node of the previous row. A naive assembly along continuous diagonals therefore couples opposite edges of the domain, which is unphysical; the corresponding entries of the first sub- and superdiagonals must be zero.
3. *Missing information.* The $(N - 2)(M - 2)$ interior equations cannot determine NM unknowns. Without boundary conditions, the homogeneous system admits every discrete harmonic function, including all constants and linear functions, as a solution. This reflects the elliptic character of the problem (Section 2.3): the solution is determined by data on the whole boundary, which the boundary conditions of Section 4 supply.

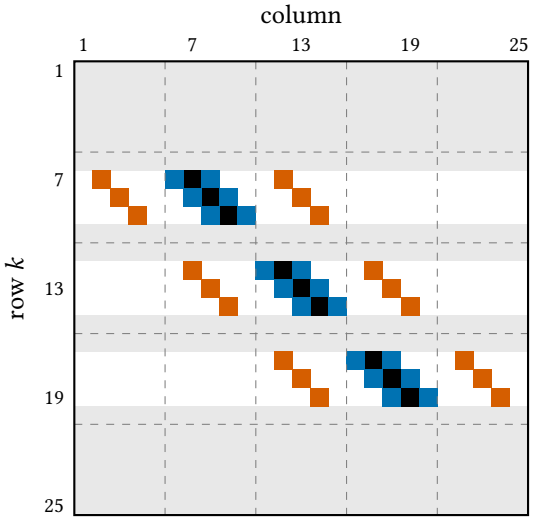


Figure 3: Sparsity pattern of the 25×25 coefficient matrix of the 5×5 mesh before boundary conditions are applied. Black: diagonal $2(1 + \lambda)$; blue: horizontal neighbours -1 at distance 1; orange: vertical neighbours $-\lambda$ at distance $N = 5$. Grey rows belong to boundary nodes, for which the stencil is incomplete. Dashed lines delimit the 5×5 blocks that correspond to mesh rows.

4 Boundary conditions

Following the script, we prescribe the concentration on the upper and lower boundaries (Dirichlet conditions $\bar{\rho}_{\text{upper}}$ and $\bar{\rho}_{\text{lower}}$) and the diffusion flux on the left and right boundaries (Neumann conditions $\bar{j}_{\text{left}}$ and $\bar{j}_{\text{right}}$). Figure 4 shows the configuration for the 5×5 mesh. The nodes on the upper and lower boundaries, labelled A to E and F to J, carry known values and are no longer unknowns. The remaining $N(M - 2) = 15$ nodes are renumbered row-wise from 1 to 15, so that node (i, j) with $2 \le j \le M - 1$ receives the number $k = (M - 1 - j)N + i$.

4.1 Dirichlet boundary conditions (Task 3.1)

> **Task 3.1.** The concentrations $\bar{\rho}_{\text{upper}}$ and $\bar{\rho}_{\text{lower}}$ are prescribed on the upper and lower boundaries, so the nodes there are no longer unknowns (Figure 4). State which nodes enter the scheme at node 3, derive the corresponding linear equation using that ρ_C is known, and collect all known quantities on the right-hand side.

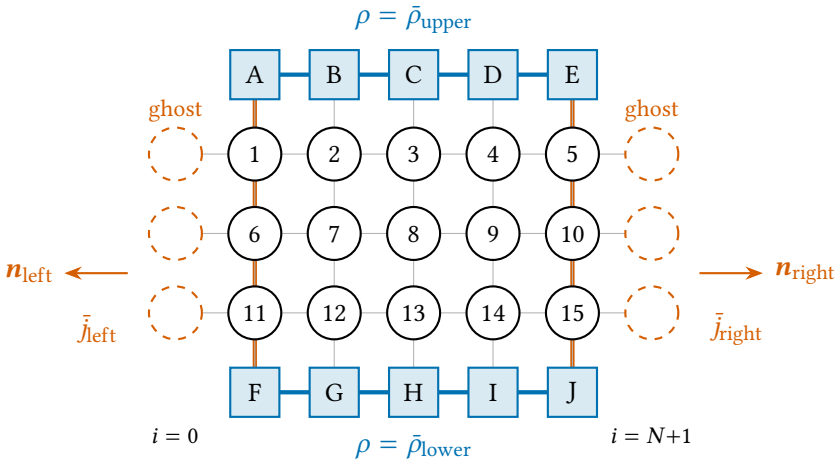


Figure 4: Boundary conditions and discretization of the 5×5 mesh. Squares (blue): Dirichlet nodes A to E ($\bar{\rho}_{\text{upper}}$) and F to J ($\bar{\rho}_{\text{lower}}$), whose values are known. Circles: the 15 unknown nodes, numbered row-wise. Dashed circles (orange): ghost nodes outside the domain, used to impose the prescribed fluxes $\bar{j}_{\text{left}}$ and $\bar{j}_{\text{right}}$ on the left and right boundaries with outward normals $\boldsymbol{n}_{\text{left}}$ and $\boldsymbol{n}_{\text{right}}$.

Node 3 is the mesh node $(i, j) = (3, 4)$ in the top row of unknowns. The stencil at this node involves its left and right neighbours 2 and 4, its lower neighbour 8 and its upper neighbour C, which is the Dirichlet node $(3, 5)$ with the known value $\rho_C = \bar{\rho}_{\text{upper}}$. Equation (3.6) at node 3 reads

$$2(1 + \lambda)\,\rho_3 - \rho_2 - \rho_4 - \lambda\,\rho_8 - \lambda\,\rho_C = 0. \tag{4.1}$$

Moving the known value to the right-hand side gives

$$\boxed{-\rho_2 + 2(1 + \lambda)\,\rho_3 - \rho_4 - \lambda\,\rho_8 = \lambda\,\bar{\rho}_{\text{upper}}.} \tag{4.2}$$

The Dirichlet node thus disappears from the unknowns and contributes $\lambda\,\bar{\rho}_{\text{upper}}$ to the right-hand side; its coefficient is removed from the row. The same holds for nodes 1 to 5 with the upper boundary and for nodes 11 to 15 with the lower boundary, where $\lambda\,\bar{\rho}_{\text{lower}}$ enters. Eliminating the Dirichlet nodes, rather than keeping them as unknowns with trivial equations $\rho_C = \bar{\rho}_{\text{upper}}$, reduces the size of the system and preserves the structure of the matrix discussed in Section 5.2.

4.2 Neumann boundary conditions (Task 3.2)

> **Task 3.2.** The diffusion fluxes $\bar{j}_{\text{left}}$ and $\bar{j}_{\text{right}}$ are prescribed on the left and right boundaries, measured along the outward normal, so that a negative value denotes flux into the domain. Discretize Fick's first law by finite differences and use it to express the ghost nodes outside the domain through known quantities, taking the direction of the flux into account. Derive the equation for node 10 with all known quantities on the right-hand side.

Prescribed flux. On the left and right boundaries the normal component of the flux, $\bar{j} = \boldsymbol{j} \cdot \boldsymbol{n}$, is prescribed, where $\boldsymbol{n}$ is the outward unit normal ($\boldsymbol{n}_{\text{left}} = -\boldsymbol{e}_x$, $\boldsymbol{n}_{\text{right}} = +\boldsymbol{e}_x$). By this convention, $\bar{j} < 0$ if the flux is directed into the domain and $\bar{j} > 0$ if it leaves the domain. With Fick's first law for an isotropic medium, $\boldsymbol{j} \cdot \boldsymbol{n} = -D\,\partial\rho/\partial n$, so that

$$\frac{\partial \rho}{\partial n} = -\frac{\bar{j}}{D} \quad \Longrightarrow \quad \partial_x \rho\big|_{\text{right}} = -\frac{\bar{j}_{\text{right}}}{D}, \qquad \partial_x \rho\big|_{\text{left}} = +\frac{\bar{j}_{\text{left}}}{D}. \tag{4.3}$$

The sign difference between the two boundaries reflects the opposite orientation of the outward normals: an outward flux at the right boundary requires a concentration that decreases towards the boundary, whereas at the left boundary it requires a concentration that increases away from it.

Ghost nodes. The boundary nodes on the left and right edges are unknowns, but their stencil reaches the nodes $(0, j)$ and $(N+1, j)$ outside the domain. We introduce these *ghost nodes* as auxiliary values and eliminate them by discretizing (4.3) with a central difference centred on the boundary node,

$$\frac{\rho_{N+1,j} - \rho_{N-1,j}}{2h_x} = -\frac{\bar{j}_{\text{right}}}{D} \quad \Longrightarrow \quad \rho_{N+1,j} = \rho_{N-1,j} - \frac{2h_x}{D}\,\bar{j}_{\text{right}}, \tag{4.4}$$

$$\frac{\rho_{2,j} - \rho_{0,j}}{2h_x} = +\frac{\bar{j}_{\text{left}}}{D} \quad \Longrightarrow \quad \rho_{0,j} = \rho_{2,j} - \frac{2h_x}{D}\,\bar{j}_{\text{left}}. \tag{4.5}$$

Both boundaries thus obey the same rule: the ghost value equals the value at the mirror node inside the domain, shifted by $-2h_x\bar{j}/D$. A central difference is accurate to $\mathcal{O}\left(h_x^2\right)$, consistent with the interior scheme (3.5). A one-sided difference such as $(\rho_{N,j} - \rho_{N-1,j})/h_x$ would introduce an $\mathcal{O}(h_x)$ error at the boundary and degrade the accuracy of the solution.

Equation for node 10. Node 10 is the mesh node $(5, 3)$ on the right boundary. Its neighbours are node 9 to the left, node 5 above, node 15 below and the ghost node $(6, 3)$ to the right. The stencil (3.6) gives

$$2(1+\lambda)\,\rho_{10} - \rho_9 - \rho_{6,3} - \lambda\,\rho_5 - \lambda\,\rho_{15} = 0. \tag{4.6}$$

Substituting the ghost value $\rho_{6,3} = \rho_9 - 2h_x\bar{j}_{\text{right}}/D$ from (4.4) and moving the known flux to the right-hand side yields

$$\boxed{-\lambda\,\rho_5 - 2\rho_9 + 2(1+\lambda)\,\rho_{10} - \lambda\,\rho_{15} = -\frac{2h_x}{D}\,\bar{j}_{\text{right}}.} \tag{4.7}$$

The coupling to the interior neighbour doubles because the ghost node is a reflected copy of node 9. The sign of the right-hand side is physically consistent: an outward flux ($\bar{j}_{\text{right}} > 0$) lowers the concentration at the boundary. The nodes 1, 5, 11 and 15 at the corners of the unknown region combine both types of boundary condition. For node 1, for example,

$$2(1+\lambda)\,\rho_1 - 2\rho_2 - \lambda\,\rho_6 = \lambda\,\bar{\rho}_{\text{upper}} - \frac{2h_x}{D}\,\bar{j}_{\text{left}}. \tag{4.8}$$

Remark 4.1. The Dirichlet conditions are essential for uniqueness. If fluxes were prescribed on the entire boundary, any constant could be added to a solution, the matrix would be singular, and a solution would exist only if the prescribed fluxes satisfied the compatibility condition $\oint_{\partial\Omega} \bar{j}\,\mathrm{d}s = 0$, which states that the net flux out of the domain vanishes in the stationary state.

5 Linear system

5.1 Assembly (Task 4.1)

> **Task 4.1.** Assemble the complete linear system $\mathbf{A}\boldsymbol{\rho} = \boldsymbol{b}_\rho - \boldsymbol{b}_j$, in which the two right-hand-side vectors carry the prescribed concentrations and fluxes. Identify which entries belong to the upper, lower, left and right boundaries and which to interior nodes.

Collecting the equations of all 15 unknown nodes, with interior rows of the form (3.9), Dirichlet contributions as in (4.2) and Neumann contributions as in (4.7), yields the linear system

$$\mathbf{A}\,\boldsymbol{\rho} = \boldsymbol{b}_\rho - \boldsymbol{b}_j, \tag{5.1}$$

where $\boldsymbol{\rho} = (\rho_1, \dots, \rho_{15})^\mathsf{T}$ is the vector of unknowns, $\boldsymbol{b}_\rho$ contains the Dirichlet data and $\boldsymbol{b}_j$ the Neumann data. For an isotropic medium on a square mesh ($\lambda = 1$, $h_x = h_y = h$), the system reads

$$\underbrace{\begin{bmatrix}
4 & -2 & \cdot & \cdot & \cdot & -1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
-1 & 4 & -1 & \cdot & \cdot & \cdot & -1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & -1 & 4 & -1 & \cdot & \cdot & \cdot & -1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & -1 & 4 & -1 & \cdot & \cdot & \cdot & -1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & -2 & 4 & \cdot & \cdot & \cdot & \cdot & -1 & \cdot & \cdot & \cdot & \cdot & \cdot \\
-1 & \cdot & \cdot & \cdot & \cdot & 4 & -2 & \cdot & \cdot & \cdot & -1 & \cdot & \cdot & \cdot & \cdot \\
\cdot & -1 & \cdot & \cdot & \cdot & -1 & 4 & -1 & \cdot & \cdot & \cdot & -1 & \cdot & \cdot & \cdot \\
\cdot & \cdot & -1 & \cdot & \cdot & \cdot & -1 & 4 & -1 & \cdot & \cdot & \cdot & -1 & \cdot & \cdot \\
\cdot & \cdot & \cdot & -1 & \cdot & \cdot & \cdot & -1 & 4 & -1 & \cdot & \cdot & \cdot & -1 & \cdot \\
\cdot & \cdot & \cdot & \cdot & -1 & \cdot & \cdot & \cdot & -2 & 4 & \cdot & \cdot & \cdot & \cdot & -1 \\
\cdot & \cdot & \cdot & \cdot & \cdot & -1 & \cdot & \cdot & \cdot & \cdot & 4 & -2 & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & -1 & \cdot & \cdot & \cdot & -1 & 4 & -1 & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & -1 & \cdot & \cdot & \cdot & -1 & 4 & -1 & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & -1 & \cdot & \cdot & \cdot & -1 & 4 & -1 \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & -1 & \cdot & \cdot & \cdot & -2 & 4
\end{bmatrix}}_{\mathbf{A}}
\begin{bmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \\ \rho_4 \\ \rho_5 \\ \rho_6 \\ \rho_7 \\ \rho_8 \\ \rho_9 \\ \rho_{10} \\ \rho_{11} \\ \rho_{12} \\ \rho_{13} \\ \rho_{14} \\ \rho_{15} \end{bmatrix}
= \underbrace{\begin{bmatrix} \bar{\rho}_{\text{upper}} \\ \bar{\rho}_{\text{upper}} \\ \bar{\rho}_{\text{upper}} \\ \bar{\rho}_{\text{upper}} \\ \bar{\rho}_{\text{upper}} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \bar{\rho}_{\text{lower}} \\ \bar{\rho}_{\text{lower}} \\ \bar{\rho}_{\text{lower}} \\ \bar{\rho}_{\text{lower}} \\ \bar{\rho}_{\text{lower}} \end{bmatrix}}_{\boldsymbol{b}_\rho}
- \frac{2h}{D} \underbrace{\begin{bmatrix} \bar{j}_{\text{left}} \\ 0 \\ 0 \\ 0 \\ \bar{j}_{\text{right}} \\ \bar{j}_{\text{left}} \\ 0 \\ 0 \\ 0 \\ \bar{j}_{\text{right}} \\ \bar{j}_{\text{left}} \\ 0 \\ 0 \\ 0 \\ \bar{j}_{\text{right}} \end{bmatrix}}_{\boldsymbol{b}_j}, \tag{5.2}$$

where dots denote zeros. For general λ and mesh spacing, the matrix has the block structure

$$\mathbf{A} = \begin{bmatrix} \mathbf{T} & -\lambda\mathbf{I} & \mathbf{0} \\ -\lambda\mathbf{I} & \mathbf{T} & -\lambda\mathbf{I} \\ \mathbf{0} & -\lambda\mathbf{I} & \mathbf{T} \end{bmatrix}, \qquad \mathbf{T} = \begin{bmatrix} d & -2 & & & \\ -1 & d & -1 & & \\ & -1 & d & -1 & \\ & & -1 & d & -1 \\ & & & -2 & d \end{bmatrix}, \qquad d := 2(1+\lambda), \tag{5.3}$$

where $\mathbf{I}$ is the 5×5 identity, and the right-hand side generalizes to $\boldsymbol{b}_\rho = \lambda\,(\bar{\rho}_{\text{upper}}\mathbf{1}_5,\ \mathbf{0}_5,\ \bar{\rho}_{\text{lower}}\mathbf{1}_5)^\mathsf{T}$ and $\boldsymbol{b}_j = (2h_x/D)\,(\bar{j}_{\text{left}}, 0, 0, 0, \bar{j}_{\text{right}}, \dots)^\mathsf{T}$, with $\mathbf{1}_5$ and $\mathbf{0}_5$ the vectors of five ones and five zeros. Each block row of $\mathbf{A}$ corresponds to one row of unknown nodes, and each block $\mathbf{T}$ couples the nodes within that row. Table 2 summarizes how the boundaries enter the system. The first column on the right-hand side, $\boldsymbol{b}_\rho$, carries the Dirichlet data of the upper and lower boundaries; the second, $\boldsymbol{b}_j$, carries the Neumann data of the left and right boundaries.

Table 2: Contribution of each boundary to the rows of the linear system (5.2).

Boundary	Rows k	Effect on the system
Upper (Dirichlet)	1 to 5	No coupling to a block above; entry $\lambda\bar{\rho}_{\text{upper}}$ in $\boldsymbol{b}_\rho$
Lower (Dirichlet)	11 to 15	No coupling to a block below; entry $\lambda\bar{\rho}_{\text{lower}}$ in $\boldsymbol{b}_\rho$
Left (Neumann)	1, 6, 11	Coefficient -2 at column $k+1$; entry $2h_x\bar{j}_{\text{left}}/D$ in $\boldsymbol{b}_j$
Right (Neumann)	5, 10, 15	Coefficient -2 at column $k-1$; entry $2h_x\bar{j}_{\text{right}}/D$ in $\boldsymbol{b}_j$
Interior	7, 8, 9	Full stencil (3.9); zero right-hand side

5.2 Properties of the coefficient matrix (Task 4.2)

> **Task 4.2.** Discuss the numerical properties of the coefficient matrix, such as its structure, and the parameters that determine its shape.

For a mesh of $N_x \times N_y$ nodes, $\mathbf{A}$ is a square matrix of dimension $n = N_x(N_y - 2)$ with the following properties.

Sparsity and band structure. Each row contains at most five non-zero entries, so $\mathbf{A}$ has at most $5n$ non-zeros out of n^2 entries. The matrix is pentadiagonal with outer diagonals at distance N_x, so its half-bandwidth equals N_x. Equivalently, it is block tridiagonal with $N_y - 2$ tridiagonal diagonal blocks $\mathbf{T}$

of size N_x and off-diagonal blocks $-\lambda\mathbf{I}$. Storage in a sparse format therefore requires $\mathcal{O}(n)$ memory, and a banded direct solver requires $\mathcal{O}\left(nN_x^2\right)$ operations, since fill-in during the factorization is confined to the band [9].

Diagonal dominance and regularity. In every row the diagonal entry equals the sum of the magnitudes of the off-diagonal entries: $2(1+\lambda) = 1+1+\lambda+\lambda$ for interior rows and $2(1+\lambda) = 2+\lambda+\lambda$ for Neumann rows. In the $2N_x$ rows adjacent to a Dirichlet boundary one off-diagonal entry has been removed, and the inequality is strict. Since the mesh is connected, $\mathbf{A}$ is irreducible. An irreducible matrix that is weakly diagonally dominant in all rows and strictly diagonally dominant in at least one row is nonsingular [8]. The discrete problem therefore has a unique solution for all boundary data.

M-matrix and monotonicity. The diagonal entries of $\mathbf{A}$ are positive and the off-diagonal entries are non-positive. Together with the irreducible diagonal dominance, this makes $\mathbf{A}$ a nonsingular M-matrix, whose inverse is entrywise non-negative [8]. Two physically meaningful consequences follow. First, non-negative boundary concentrations combined with non-positive (inward) fluxes give $\boldsymbol{b}_\rho - \boldsymbol{b}_j \geq 0$ and hence a non-negative concentration everywhere. Second, the solution depends monotonically on the data: raising a boundary concentration never lowers the concentration at any node. This is the discrete counterpart of the maximum principle.

Symmetry and spectrum. The matrix is not symmetric, because each Neumann row couples to its interior neighbour with the coefficient -2, while the interior neighbour couples back with -1 (for example $a_{1,2} = -2$ but $a_{2,1} = -1$). The asymmetry is removed by halving the Neumann rows: with $\mathbf{W} = \mathrm{diag}(w_k)$, where $w_k = 1/2$ for k on the left or right boundary and $w_k = 1$ otherwise, the matrix $\mathbf{S} = \mathbf{WA}$ is symmetric. As a symmetric, diagonally dominant matrix with positive diagonal, $\mathbf{S}$ is positive semidefinite by Gershgorin's theorem, and since it is nonsingular, it is positive definite. Because $\mathbf{A} = \mathbf{W}^{-1}\mathbf{S}$ is similar to the symmetric positive definite matrix $\mathbf{W}^{-1/2}\mathbf{S}\,\mathbf{W}^{-1/2}$, all eigenvalues of $\mathbf{A}$ are real and positive. The symmetrized system $\mathbf{S}\boldsymbol{\rho} = \mathbf{W}(\boldsymbol{b}_\rho - \boldsymbol{b}_j)$ can therefore be solved with the Cholesky factorization or the conjugate gradient method.

Conditioning. The smallest eigenvalue of $\mathbf{A}$ scales as $\mathcal{O}\left(h^2\right)$ and the largest remains bounded, so that the spectral condition number grows as $\mathcal{O}\left(h^{-2}\right)$ under mesh refinement [7]. Fine meshes thus require more iterations of iterative solvers, and round-off errors are amplified more strongly.

Parameters that influence the shape of the matrix.

- The number of nodes per row, N_x, sets the size of the blocks and the distance of the outer diagonals from the main diagonal, that is, the bandwidth.
- The number of rows, N_y, sets the number of block rows, which is $N_y - 2$ after elimination of the Dirichlet nodes.
- The numbering scheme determines which dimension sets the bandwidth. Column-wise numbering would yield the half-bandwidth $N_y - 2$ instead of N_x, so the cost of a banded solver is minimized by numbering along the shorter side of the mesh.
- The type of boundary condition modifies the boundary rows: Dirichlet conditions remove rows and columns and make the adjacent rows strictly diagonally dominant; Neumann conditions double the coupling to the interior neighbour and break the symmetry.
- The weight $\lambda = Ph_x^2/h_y^2$, which depends on the anisotropy and on the domain size and node numbers through h_x and h_y, changes the values of the entries but not the sparsity pattern.
- If the principal axes of $\mathbf{D}$ were not aligned with the mesh, the mixed derivative in (2.6) would require the diagonal neighbours $(i \pm 1, j \pm 1)$, giving a nine-point stencil with four additional diagonals at distances $N_x \pm 1$.

6 Summary

We have derived Fick's second law for an anisotropic two-dimensional medium from the continuity equation and Fick's first law, and shown that it is parabolic, while its stationary limit is elliptic. In the principal axes of the diffusion tensor, stationary diffusion obeys $\partial^2_{xx}\rho + P\,\partial^2_{yy}\rho = 0$, where P is the ratio of the principal diffusivities, and a rescaling of one coordinate maps this equation onto the Laplace equation. A second-order central-difference scheme on an equidistant mesh, combined with row-wise numbering, elimination of Dirichlet nodes and ghost nodes for the Neumann conditions, leads to the linear system (5.1). Its coefficient matrix is sparse, pentadiagonal and block tridiagonal, irreducibly diagonally dominant and a nonsingular M-matrix with real positive eigenvalues; the asymmetry caused by the Neumann rows is removed by a diagonal scaling. These properties guarantee a unique discrete solution that is non-negative for physically admissible data and depends monotonically on them, and they permit efficient sparse direct or iterative solvers. Part II implements this system for arbitrary domain sizes, node numbers and boundary values, and verifies it against the test problems of Task 5.

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